Abdelrahman Tayel

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Education

Poznań University of Technology — Poznań, Poland
B.Sc. Automatic Control and Robotics — Oct 2023 – Apr 2027
Thesis (in progress): Modular Hardware Architecture for Autonomous Mobile Robots

Experience

Elsewedy Industries (EMAS) — Cairo, Egypt
Automation Engineer Intern — Aug 2026 – Sep 2026

- Prepared technical and commercial offers for PLC, RTU, and substation automation packages
- Reviewed P&ID, electrical schematics, and panel drawings; wrote compliance sheets for client specifications
- Selected PLC/RTU architectures for tenders, evaluating I/O, network topology, and redundancy

Memico — Cairo, Egypt
Electrical Power Engineering Intern — Jul 2026 – Aug 2026

- Followed assembly and Factory Acceptance Testing (FAT) of LV/MV switchgear and distribution panels
- Reviewed QC checklists; read busbar layouts and wiring schematics in EPLAN,and AutoCAD

Poznań University of Technology — Poznań, Poland
Research Assistant Intern — Jun 2025 – Aug 2025

- Ran fault-injection and hardware-in-the-loop tests on PMSM and BLDC drives using DSP-based hardware
- Controlled a BLDC motor on STM32 Nucleo (GPIO, timers, ADC)
- Wrote MATLAB and Python scripts to analyze 3-phase voltage, current, and speed data

Projects

Modular Hardware Platform for Autonomous Mobile Robots — In Progress

- Designing multi-board PCB architecture with a protected power distribution board (dual-source switching, reverse-polarity and overcurrent protection, 24 V / 5 V / 3.3 V rails)
- Developing BLDC drive units with closed-loop control and current, voltage, and thermal telemetry
- Validating the platform through electrical load testing, thermal profiling, and drive-cycle benchmarks
- Designed a custom 3D-printed chassis for modular serviceability and wiring harness routing

Siemens S7-1200 Automation Stations — May 2026

- Built a sorting belt, multi-floor elevator (SCL/LAD), and pump station with HMI over Profinet
- Implemented PID control with anti-windup and analog signal scaling

Skills

Technical: PCB design, C/C++, Python, MATLAB/Simulink, ROS2 (simulation), YOLOv8/OpenCV, STM32, ESP32, Git, AutoCAD, Linux (Ubuntu), Siemens TIA Portal, SCL/LAD, S7-1200, PID tuning
Language: Arabic (native), English (C1), German (A1), Polish (A1)
Laboratory: Soldering, oscilloscope debugging, wiring, electrical schematics, safety procedures, Multimeters, Converters, inverters
Interests: PCB design, Embedded systems, Robotics, Industrial Automation